

Code Worth Writing

Ray Myers, The Coding Agency

Windows

An error has occurred. To continue:

Press Enter to return to Windows, or

Press CTRL+ALT+DEL to restart your computer. If you do this,
you will lose any unsaved information in all open applications.

Error: 0E : 016F : BFF9B3D4

Press any key to continue _

DEATH

Blue screen of death

Software Should Work

Ray Myers

20 years in software

Tech Lead in 4 industries

raymyers.org

thecodingagency.org

Former:



The Indeed logo, featuring the word "indeed" in a blue, lowercase, sans-serif font. A blue arc is positioned above the "i".

The UnitedHealth Group logo, featuring the words "UnitedHealth Group" in a blue, serif font. A registered trademark symbol (®) is located at the end of the word "Group".

The OpenHands logo, featuring two yellow hands with fingers spread, one above the other, with three small black lines above the top hand. To the right of the hands, the word "OpenHands" is written in a bold, black, sans-serif font.

Software Should Work

Works as intended

Works for us

AI

Software quality crisis

Not just IT's problem

Everyone's problem

Losing public trust

When is it a crisis?

Your hair stylist knows about it

When they think of bugs, they think of AI

Unaided awareness



We beat the blue screen of death

How?

We beat BSOD with ...

- Better documentation?

We beat BSOD with ...

- Better documentation?
- Architecture?

We beat BSOD with ...

- Better documentation?
- Architecture?
- Static analysis?

Yes

Context: Device drivers, memory bugs and API changes

- Better documentation
- Architecture
- Static analysis
- Error reporting

Static analysis

- SLAM - engine for driver verification
- SLAyer - memory safety checker for C
- Terminator - proves programs will halt

“

Things like even software verification, this has been the **Holy Grail** of computer science but now in some very key areas, for example, driver verification we're building tools that can do actual **proof about the software**.

Bill Gates, WinHec 2002 keynote



Extreme measures

Behaviors are forced to change

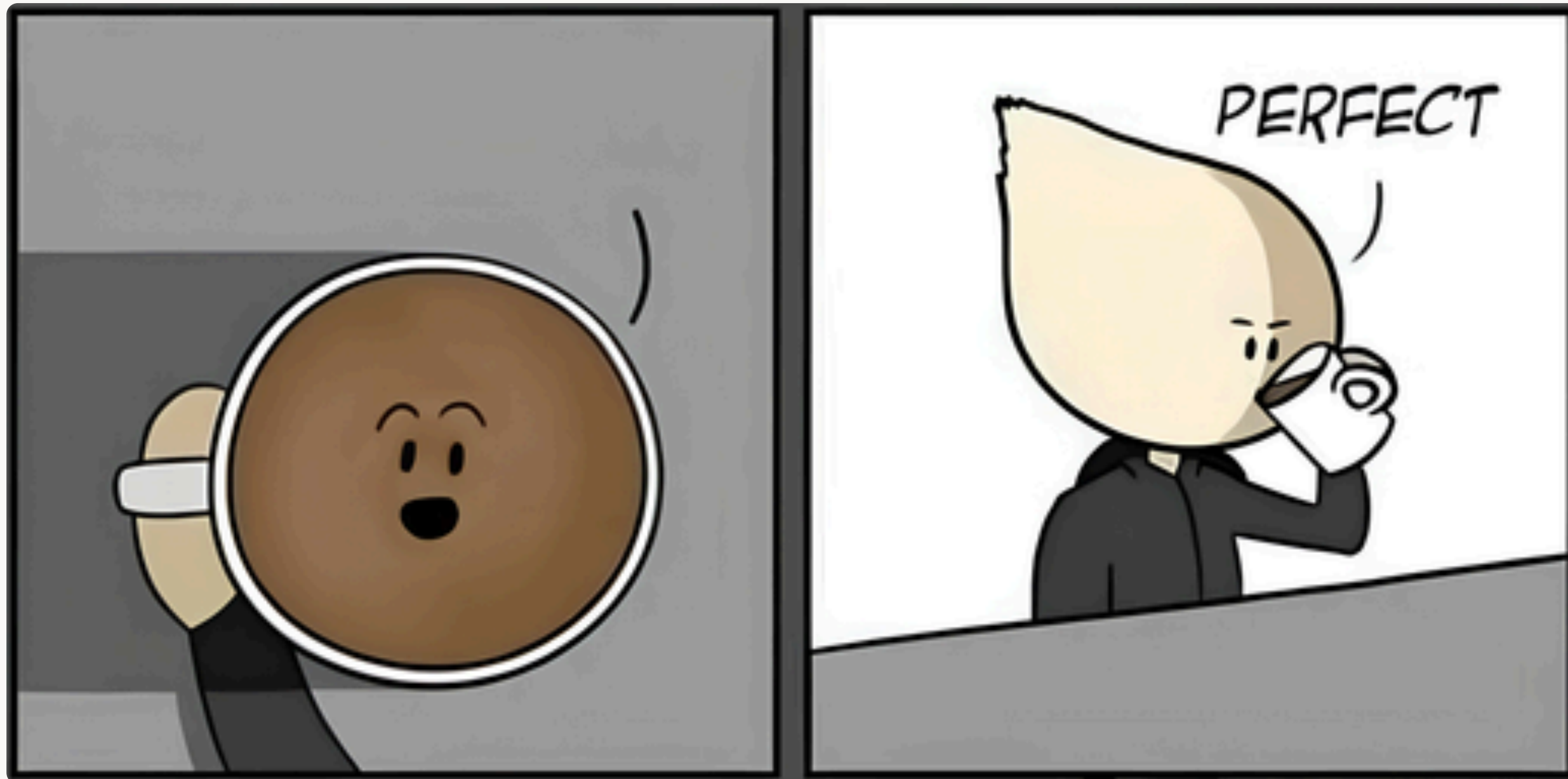
AI is transforming software

Anthropic is deceptively campaigning to end Software Engineering to support a trillion dollar IPO





AI is transforming software!



AI is transforming software
...into what?



We have agency

Programming language
designers

Predicting the future

Code is for people

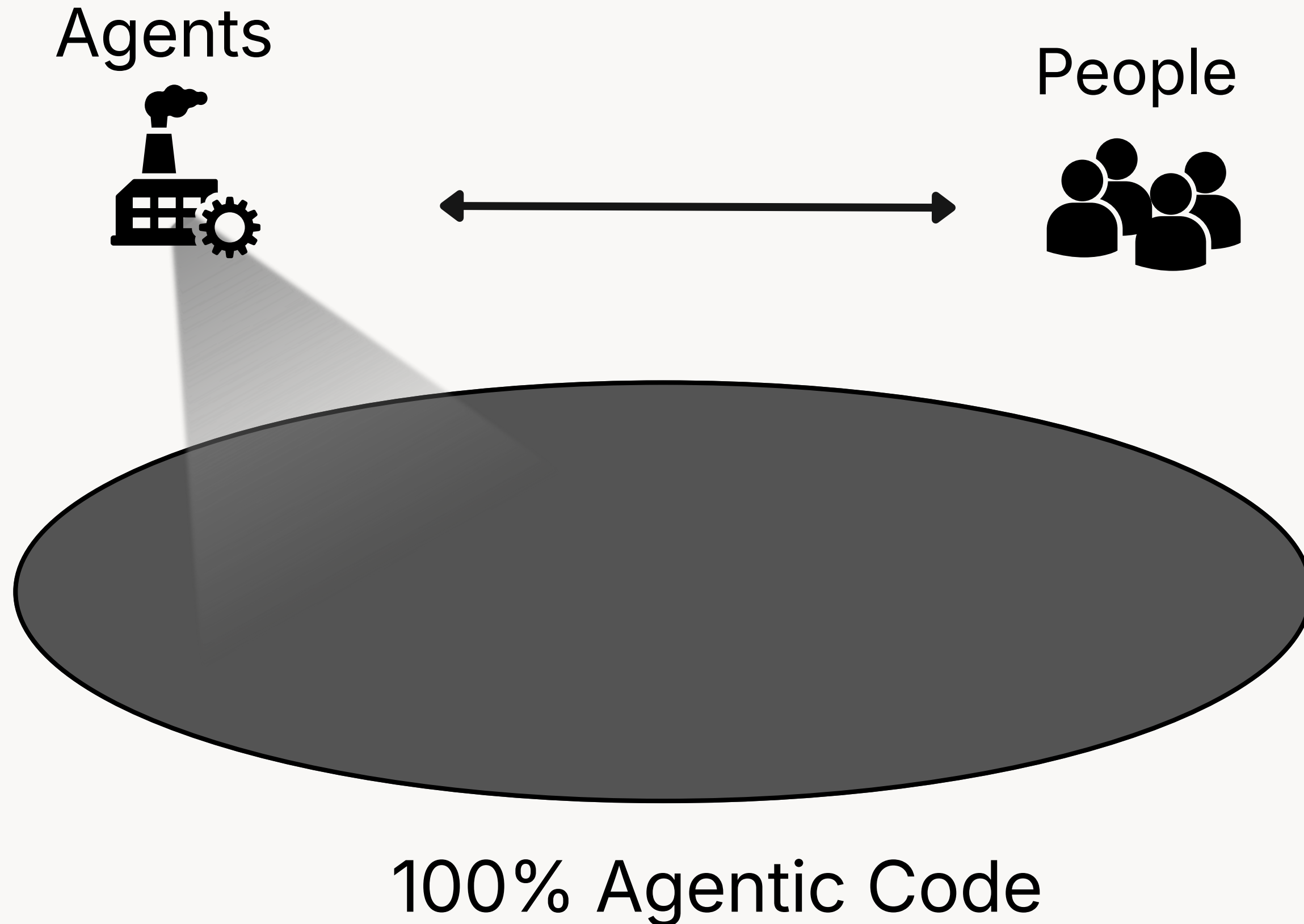
vs

Code is for agents

Agent-Maxxing



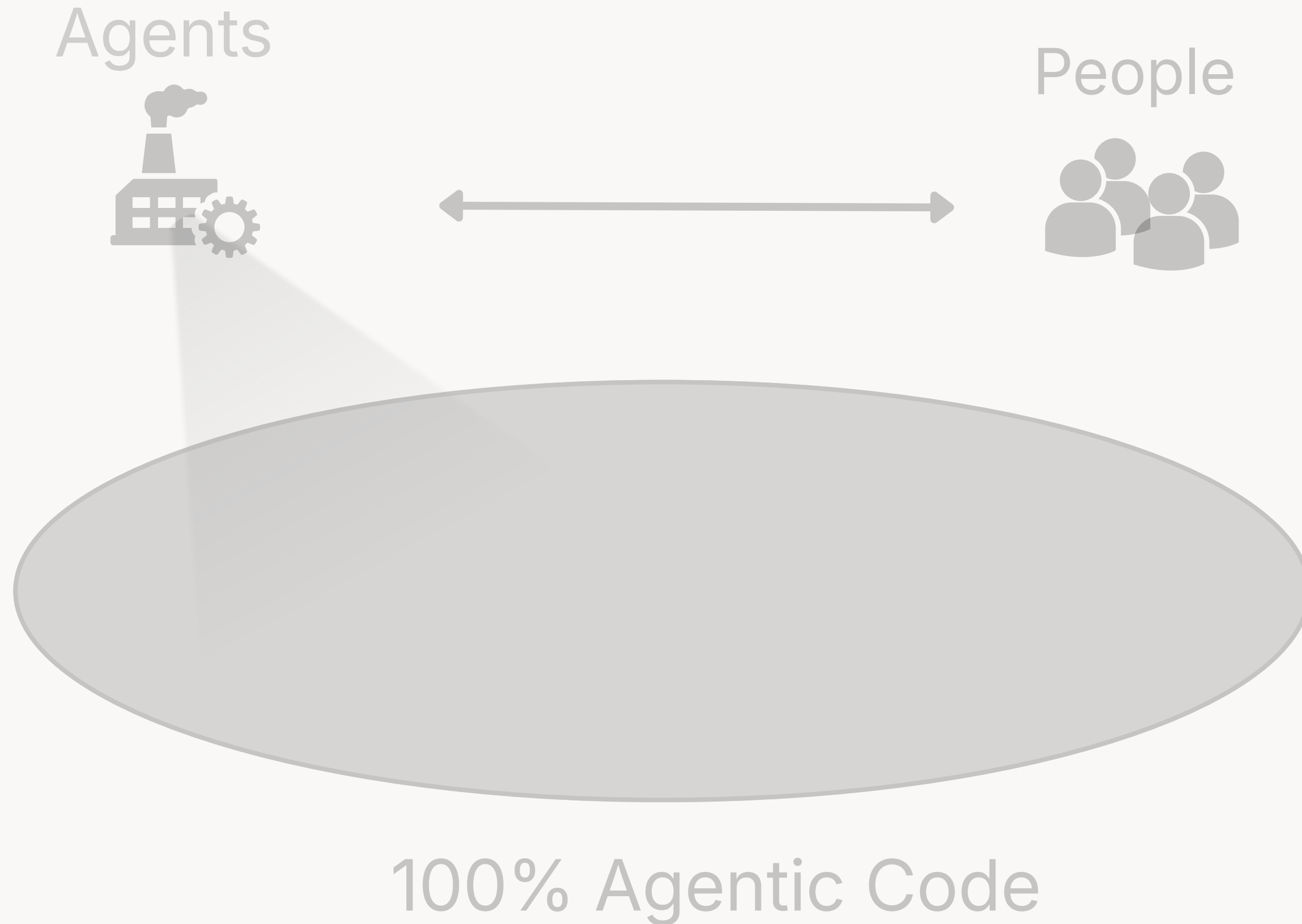
Dark Software Factory



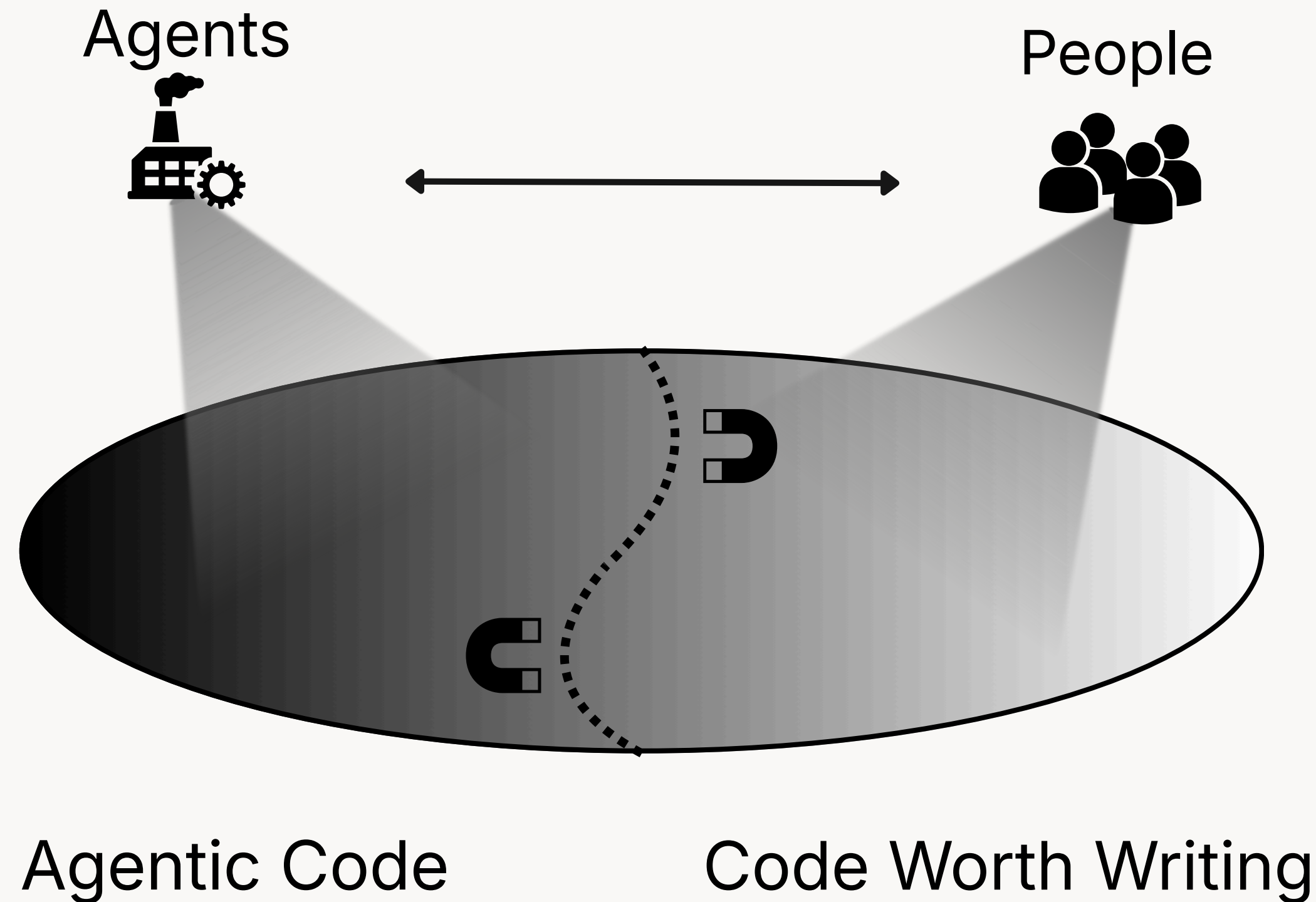
WELCOME TO
MISSOURI

THE SHOW-ME STATE

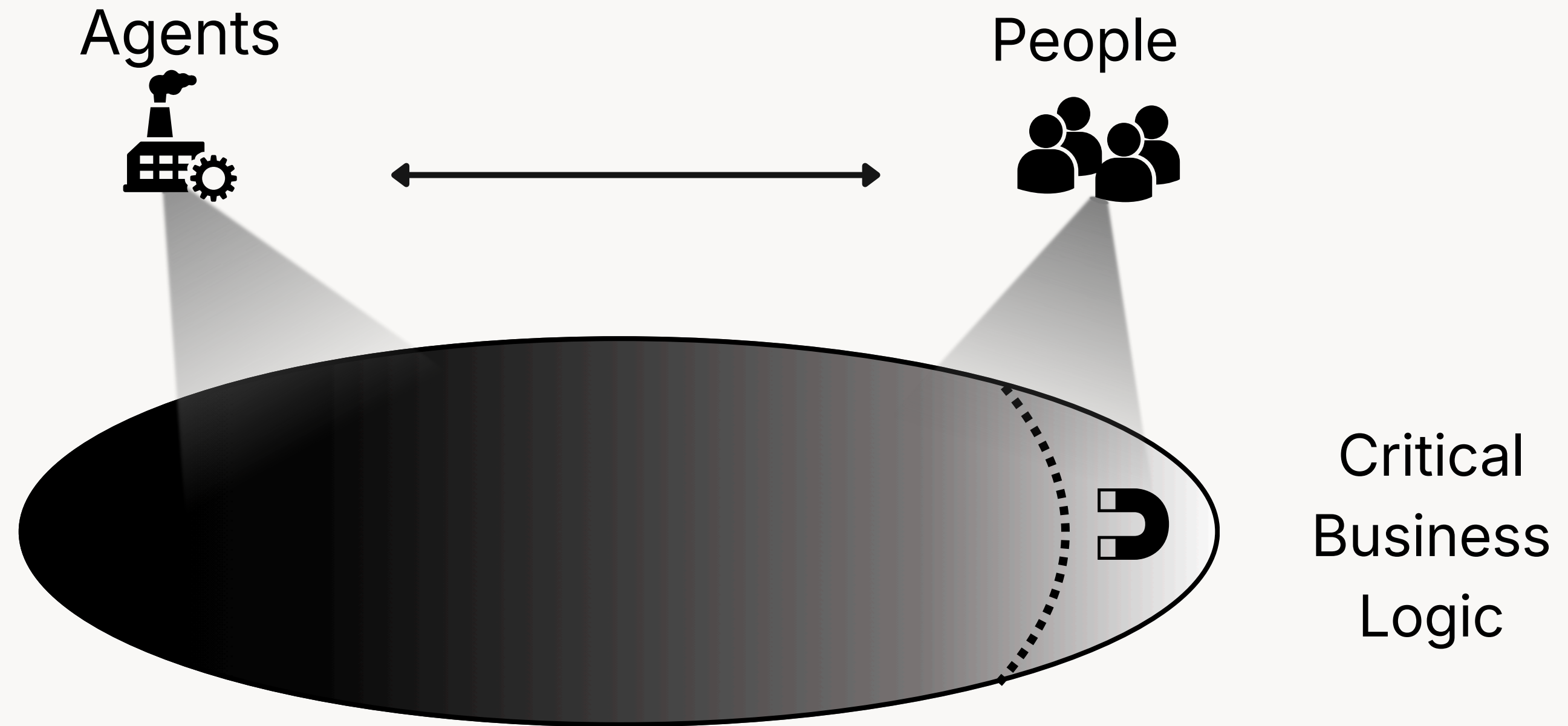
Dark Software Factory



The Show-Me Method

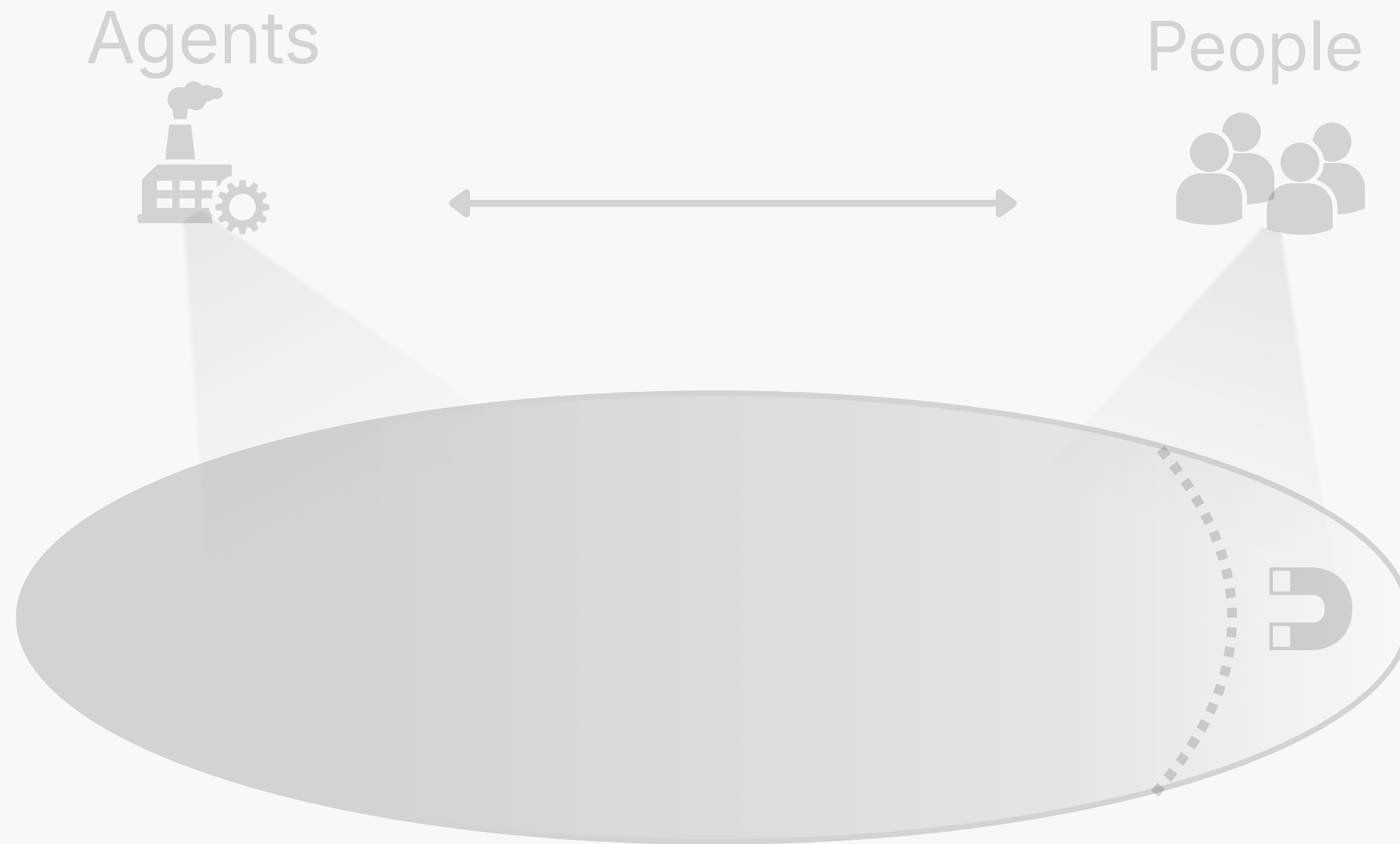


The Show-Me Method

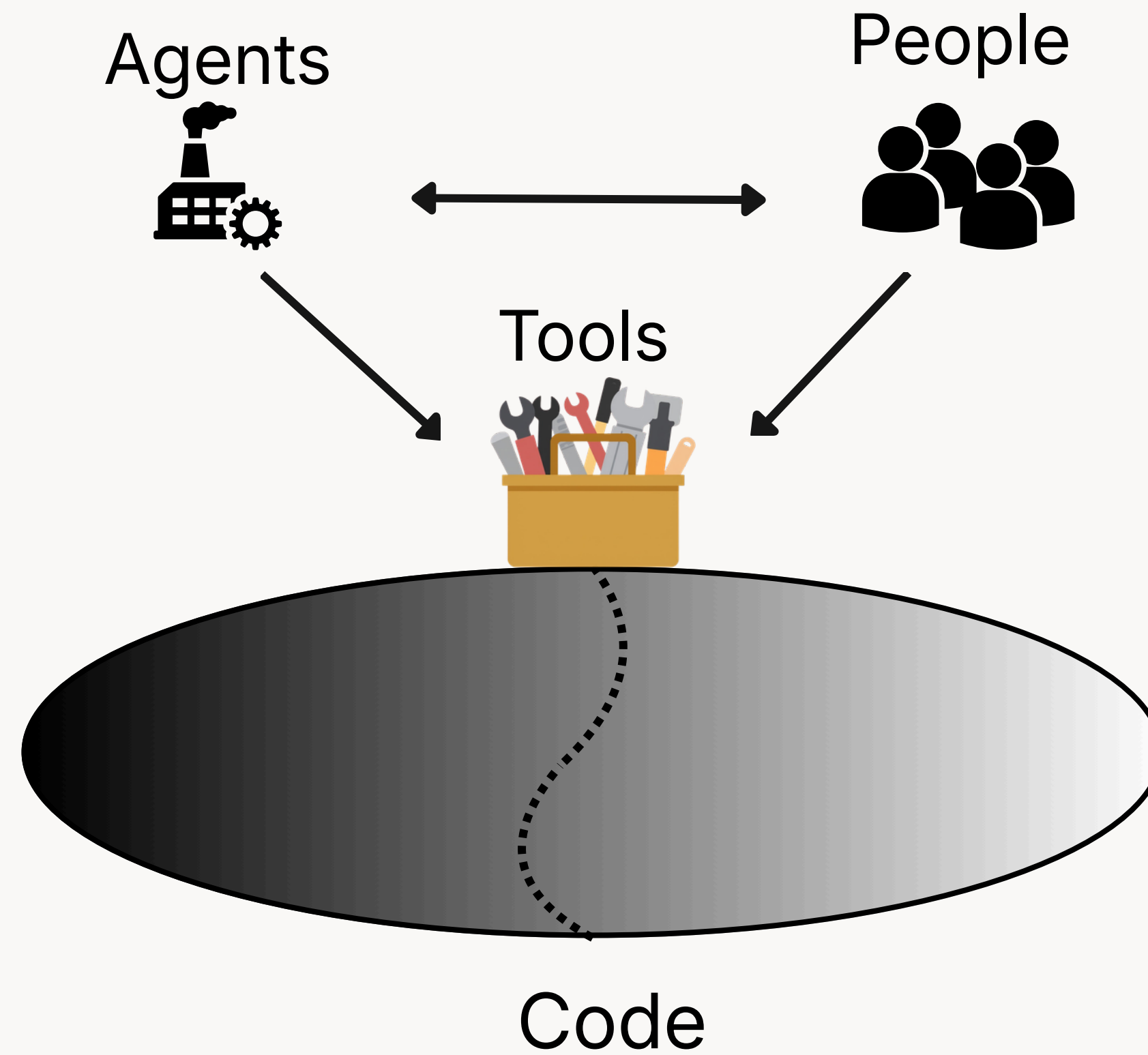


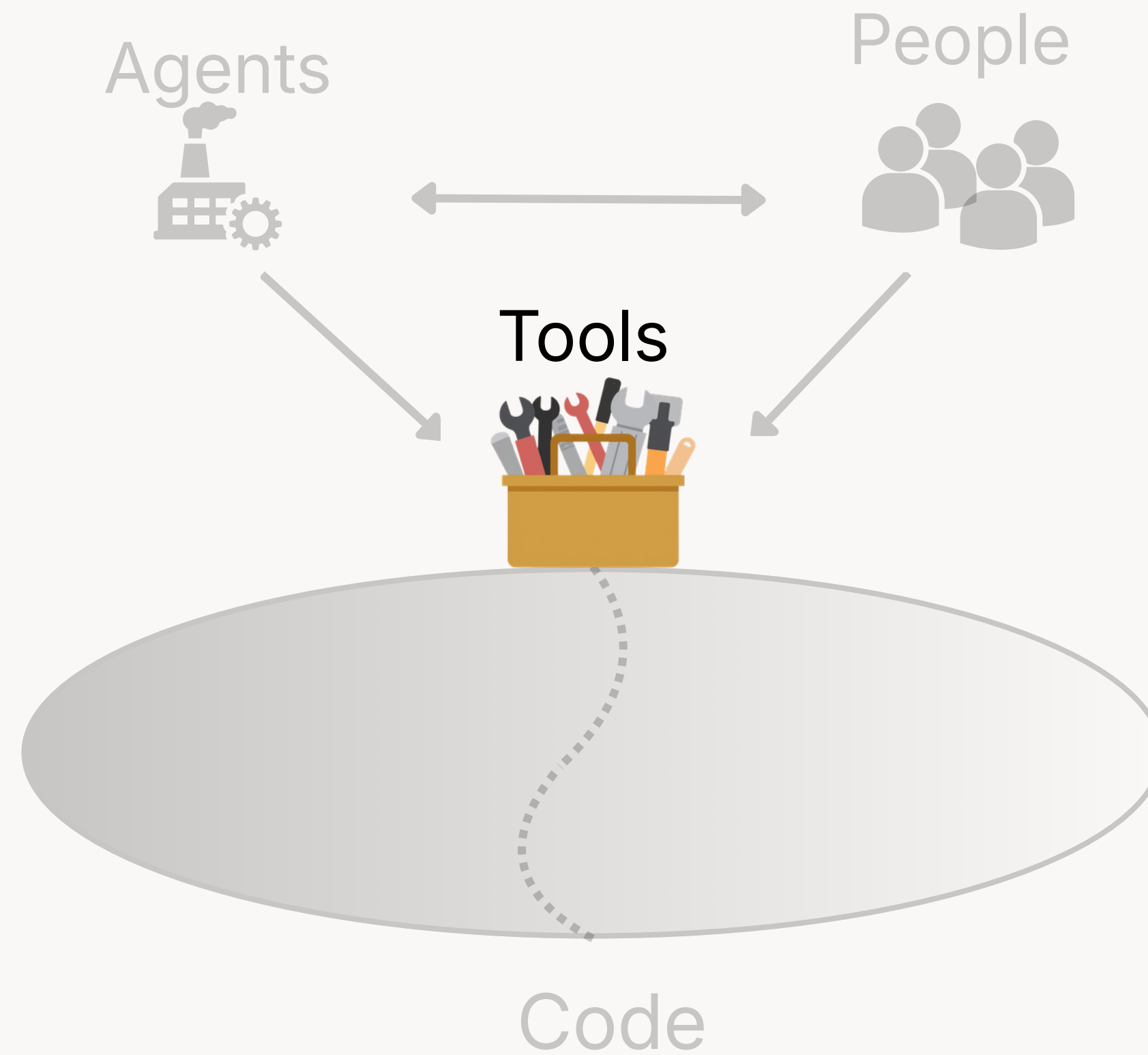
Choose your level of agentic code

The Show-Me Method



How to improve?





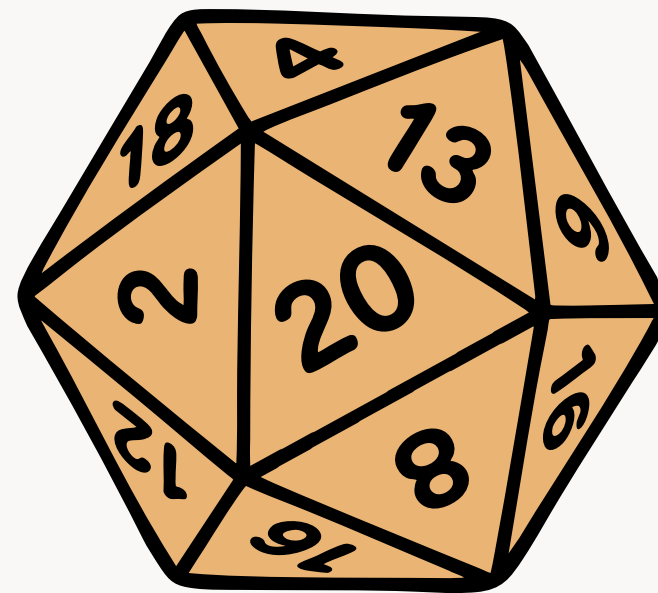
Same tools, different interfaces

LSPs → MCPs

Language Server Protocol used by IDEs

Model Context Protocol used by coding agents

The bet that can't lose



Programs that reason about programs

Tools

Compilers, Linters, IDEs,
Refactoring, Static analysis, ...

Programming languages
can enable tools

How to enable better tools?

Formal definitions



Dear Santa,



Language maintainers:

Language maintainers:

Define your syntax tree

Language maintainers:

Define your syntax tree

Define your semantics

Language maintainers:

Define your syntax tree

Define your semantics

Work with legacy code

Simplicity can help

Restrictions can help

Showcase

Showcase: Roc

```
print_remaining! = |todos|
  todos
    .keep_if(|todo| todo.status != Done)
    .for_each!(|todo| echo!("- ${todo.name}\n"))

main! = |_args| {
  todos = [
    { name: "Learn Roc",      status: Done },
    { name: "Buy groceries",  status: Done },
    { name: "Write blog post", status: InProgress },
    { name: "Call mom",       status: NotStarted },
  ]
  print_remaining!(todos)
  Ok({})
}
```



Showcase: Roc

“Fast, Friendly, Functional”

Platforms: Legacy Friendly, Restrictions

Software Unscripted Podcast

Rewritten Rust → Zig



Showcase: EYG

```
let string = @standard:1.string
let ask = perform Prompt("What is your name")
let name = match ask {
  Ok (name) -> name
  Error (_) -> "World"
}
let greeting = string.append("Hello, ", name)
perform Alert(greeting)
```

No action bound for key 'Shift'



Showcase: EYG

Defines an stable IR

Legacy friendly: embeddable, effects

Fanatically simple

Type Soundness



Type Soundness

If type checking succeeds,
programs never crash

All error cases are handled

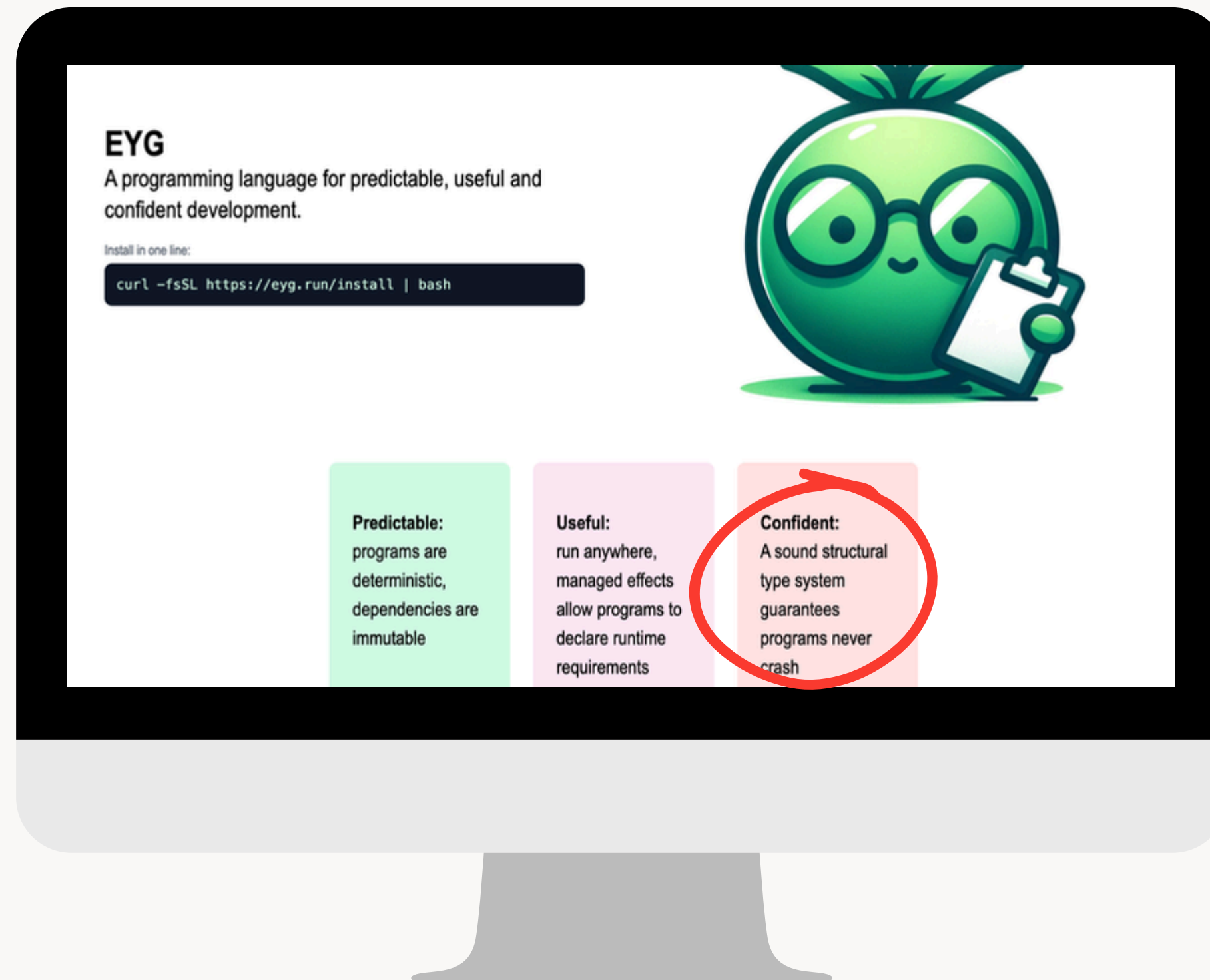
Type Soundness

If type checking succeeds,
programs never crash

All error cases are handled

Mathematical claim

Front Page Feature



`Eyg :: TypeSoundLanguage`

Gleam can't do this



We're gonna need a
bigger type checker

LENN

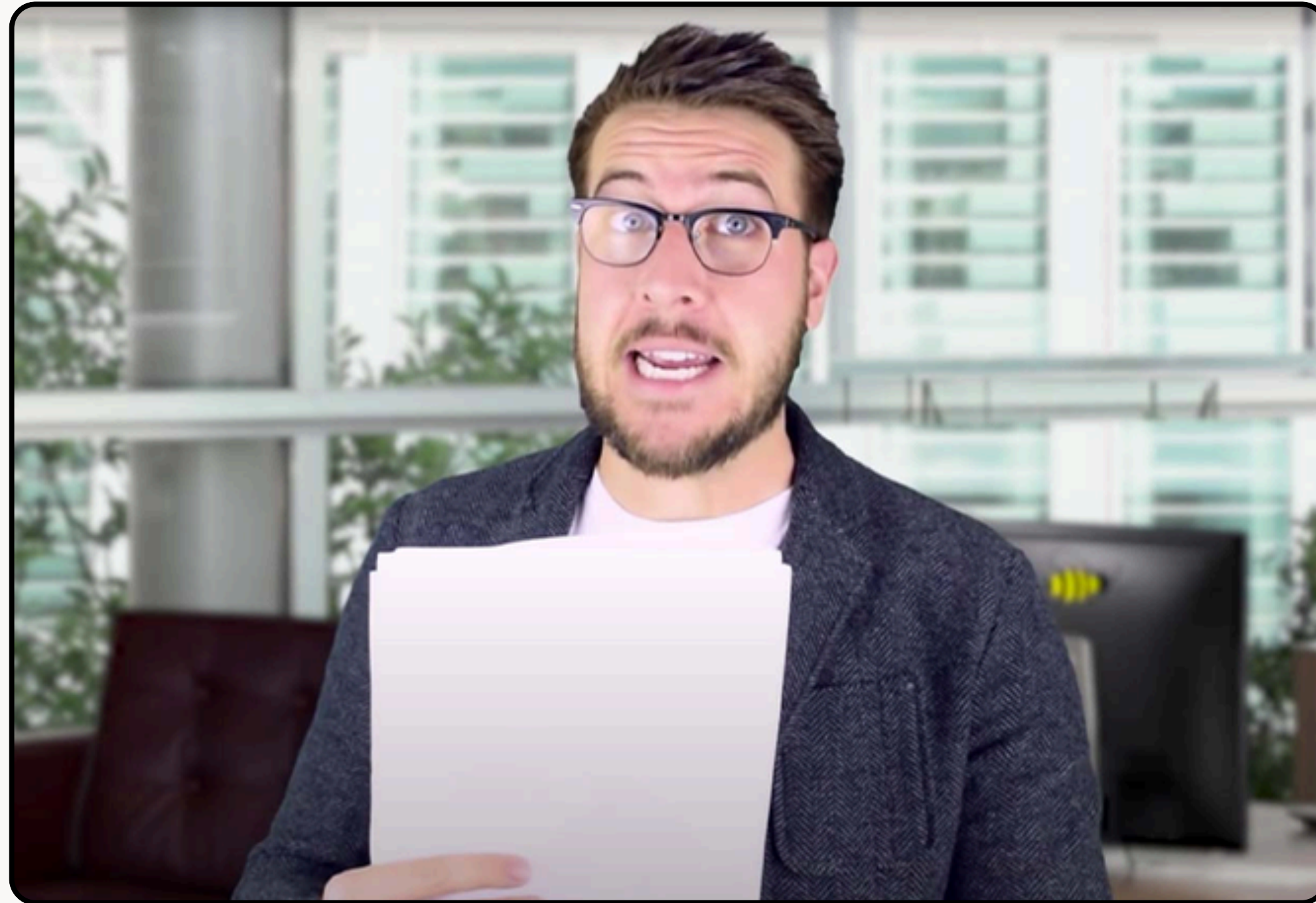
THEOREM PROVER

$$\nexists p \in \text{Program}. \text{Infer}(p) \neq \text{Error} \wedge \text{Eval}(p) = \text{Error}$$

or

$$\forall p. \text{Infer}(p) \neq \text{Error} \implies \text{Eval}(p) \neq \text{Error}$$

- Port interpreter from Gleam to Lean
- Port the type checker
- Define operational semantics and type rules
- Prove preservation and progress theorems



Super easy, barely an
inconvenience

Snippet: Functional Big-Step Semantics

```
-- The fuel-indexed evaluator. One 'step' per unit of fuel; 'eval 0 = timeout'.
Reuses the interpreter's 'step', so the reduction rules are shared. -/
def eval [BEq m] : Nat → Config m → Result m
| 0, _ ⇒ .timeout
| fuel + 1, (c, e, k) ⇒
  match step c e k with
  | .Loop c' e' k' ⇒ eval fuel (c', e', k')
  | .Break (.ok v) ⇒ .done (.value v)
  | .Break (.error (.UnhandledEffect op lift, _, env, k')) ⇒
    .effect op lift (fun reply ⇒ (.V reply, env, k'))
  | .Break (.error (reason, _, _, _)) ⇒ .done (.crash reason)
```

EYG is sound in theory

Not in practice, *yet*

```
> /type !fix((x) -> { !int_add(x, 1) })
Integer
```



```
> !fix((x) -> { !int_add(x, 1) })
error: unexpected term, expected: Integer got: Partial(Builtin2("fixed"), fn(x) { ... })
hint: check the value passed to this operation has the expected shape
```

```
→ in <repl>:1
  1 | !fix((x) -> { !int_add(x, 1) })
      ^^^^^^^^^^^^^^^^^
in <repl>:1 (Partial(Builtin2("fixed"), fn(x) { ... })))
```

Expected behavior: The type analyzer should reject this code, consistent with the eval.

Reported: GitHub CrowdHailer/eyg-lang issue #97

Formal Verification can tell us about
behavior for all possible inputs

Start by defining your semantics

Showcase: ZIG

Defines grammar and AST library

Interop with C, C++

"Maintain it with Zig" slogan

“

Focus on debugging your application rather than debugging your programming language knowledge.

Andrew Kelly



Showcase:  ZIG

Major Zig codebases include:

TigerBeetle, Ghostty, and Bun



[AI](#)[Security](#)[Microsoft](#)[AWS](#)[Developer](#)[Open Source](#)[Columnists](#)[BOFH](#)[Who, Me?](#)[On Call](#)

DEVOPS

Anthropic's Bun Rust rewrite merged at speed of AI

Version 1.3.14 of JavaScript toolkit released as last Zig version; a million lines of Rust code merged in gargantuan commit

 [Tim Anderson](#)

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Published Thu 14 May 2026 // 13:01 UTC



A pull request with a Rust version of Anthropic's Bun, a JavaScript toolkit and runtime originally written in Zig, has been merged to the main Bun repository. This comes just days after its author, Jared Sumner, said "there's a very high chance all this code gets thrown out."

Sumner posted on X (formerly Twitter) five days ago that "99.8 percent of bun's pre-existing test suite passes on Linux x64f glibc in the rust rewrite," a clue that what was initially described as an experiment was likely to make it to production. Three days later, the Bun team released version 1.3.14, with Sumner stating that if the Rust rewrite was merged, "this would be the last version in Zig."



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Retrofuturist Software Mender

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Zig Creator Calls Spade a Spade, Anthropic Blows Smoke

Jul 12, 2026

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I harass the sea with my tiny boat and am called a pirate, you do it with a great fleet and are called a king.

Anthropic is actively campaigning to end software engineering. They need you to believe they can do that. Well, maybe it's not you that they need to convince. Maybe it's your C-Suite, various world leaders, or the manager of your retirement fund. They've raised \$132 billion in investment, and are approaching an IPO valued over \$1 trillion. Since they cannot show profitability, this depends on selling their hypothetical future impact.

In literary terms, Anthropic is an *unreliable narrator*.



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1. ▲ Zig Creator Calls Spade a Spade, Anthropic Blows Smoke (raymyers.org)
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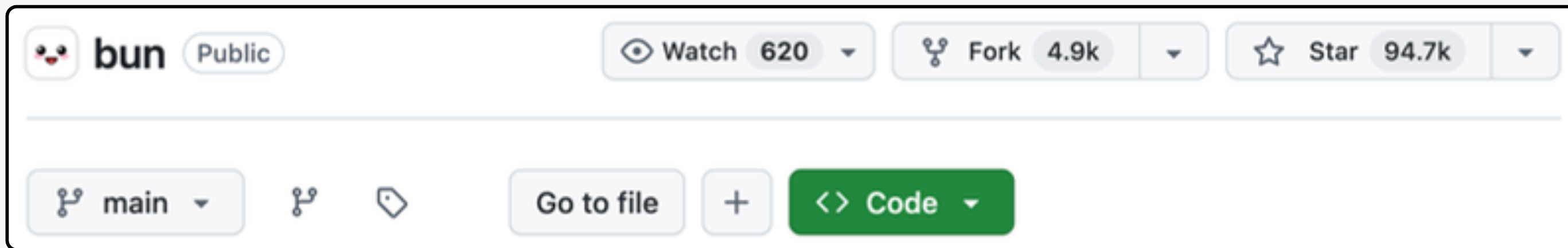
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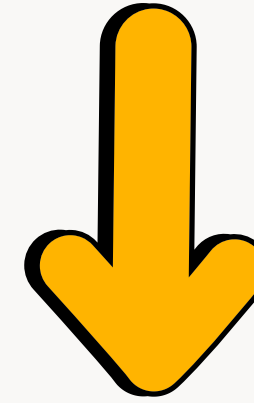
Anthropic's non-explanation,
two months late

Unstated business reasons

Could they have make Zig
work?

We'll never know





 **bun** Public

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 main ▼

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 Code ▼

 raymyers /  **bunsen** 

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forked from [oven-sh/bun](#)

 main 

 **4539** Branches  **257** Tags

Go to file 

  **Code** 

This branch is **908 commits behind** oven-sh/bun:main .

 **Contribute** 

 **Sync fork** 

 **Jarred-Sumner** Bun.serve: rename h3/h1 options to http3...  0d9b296 · 2 months ago  **15,343 Commits**

Project Bunsen

Keeping Bun in Zig

Do what Anthropic couldn't:
Debug their codebase





KILL IT WITH FIRE



MANAGE AGING COMPUTER SYSTEMS
(AND FUTURE PROOF MODERN ONES)

MARIANNE BELLOTTI



Challenge

Find 3 months worth of memory bugs in Bun in a week

One dev, one agent, no experience with the codebase

Learned Zig yesterday

Process

Investigated commits from Feb 12 to May 12, 2026

Last 3 months of Bun commits before rewrite

Surveyed 48 memory bug fixes



Process

Prioritized high-churn areas of the code

Derived conventions from past fixes

Wrote custom linters using Zig Ast library

Fuzzed with Asan and Tsan analysis



Caught 50 existing memory bugs



Caught 50 existing memory bugs

Why didn't Anthropic?



DARPA TRACTOR 2026 Report

(Translating All C To Rust)

Why not cite prior work?

Anthropic admitted AI wasn't enough

We need the language toolchain

Let's talk

We have a choice

Future of vibes

Future we define

Thank you!

raymyers.org

thecodingagency.org